

Introduction to Calculus

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6G4Z3006 Calculus

Tutorial Week 01

- Let's look at the module's [Moodle area](#).

- Your calculator/computer will tell you that $\sqrt{5}$ is

$$\sqrt{5} = 2.236067977 \dots$$

- The decimal expansion will go on forever, any finite decimal expansion can only ever be an approximation to $\sqrt{5}$.
- $\sqrt{5}$ is an **irrational** number, i.e. it cannot be given as a ratio of integers, a/b , where $a, b \in \mathbb{Z}$.
- In fact, we can show (see the level 5 Number Theory and Abstract Algebra unit) that for every integer m , its square root $\sqrt{m}$ will either be an integer itself, or if not, it will be irrational.
- However, it's also true that any irrational number can be approximate arbitrarily closely by rationals.
- But in general, to get better and better rational approximations to an irrational, we will need larger and larger denominators.

- In this tutorial we will get a first look at the important concept of **convergence of sequences**.
- We'll also see some examples of **proof by induction**.
- Don't worry if these concepts are new and unfamiliar to you. This is a *first look*.

From the APEX Calculus textbook

- Let's quickly read about [Newton's method](#) for finding roots/zeros of functions.

- We want to approximate $\sqrt{5}$, so we shall consider a function f , defined by,

$$f(x) = x^2 - 5,$$

with an initial approximation of $x_0 = 3$.

- This leads to an **infinite sequence** of values

$$x_0, x_1, x_2, x_3, \dots,$$

with $x_0 = 3$, satisfying the **recurrence relation**

$$x_{n+1} = x_n - \frac{x_n^2 - 5}{2x_n}.$$

- This sequence *should* converge to $\sqrt{5}$, i.e.

$$\lim_{n \rightarrow \infty} x_n = \sqrt{5}.$$

In this tutorial we will work towards a proof of this claim about this sequence.

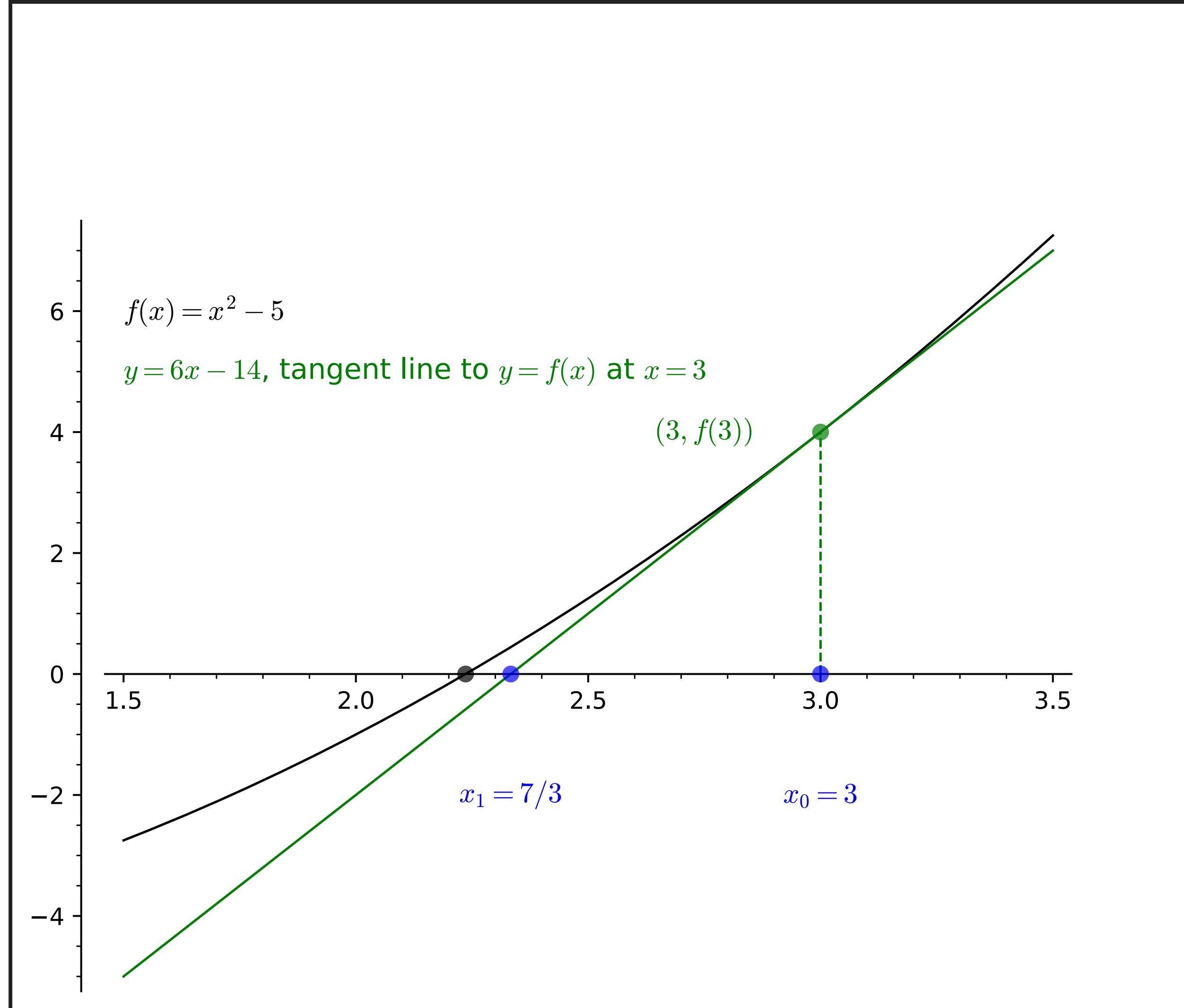
- **Task 1.** Use Newton's recurrence relation

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

to derive our recurrence relation

$$x_{n+1} = x_n - \frac{x_n^2 - 5}{2x_n}.$$

- **Task 2.** Confirm the equation of the first tangent line and confirm the value of $x_1 = 7/3$.



- **Task 3.** Take our recurrence relation and use it to show that for all $m \geq 1$ we have $x_m > 0$.
- Hints:
 - Rewrite the recurrence in the form

$$x_{n+1} = \frac{1}{2x_n} \left(\dots \right)$$

- Then construct an argument that if $x_n > 0$ then $x_{n+1} > 0$.
 - We can then use *proof by induction* to prove that the sequence is always positive.
- **Task 4.** Now let's improve the lower bound by showing that for all $m \geq 1$ we have $x_m > \sqrt{5}$.
- Hint:
 - Use the recurrence relation to find a nice expression for x_{n+1}^2 of the form

$$x_{n+1}^2 = 5 + \left(\dots \right)^2$$

- Use proof by induction to prove that for all $m \geq 1$ we have $x_m^2 > 5$.
 - Then use the result of Task 3 to deduce that in fact $x_m > \sqrt{5}$.

Proving the sequence is decreasing

- **Task 5.** Now prove the sequence is decreasing, i.e. for all $m \geq 1$ we have $x_m > x_{m+1}$.
- Hint:
 - Use the recurrence relation to get a nice expression for the difference $x_{n+1} - x_n$.
 - Use results of earlier tasks to get the decreasing conclusion.
- **Calling on some theory:** For the next step we'll need to rely on a theorem about the convergence of monotone (increasing or decreasing) bounded sequences. This is [Theorem 9.1.32 in APEX Calculus](#).
- So now we know our sequence is convergent, i.e. it has a limit. Newton's method claims that this limit should be $\sqrt{5}$.
- **Task 6.** Prove that our sequence converges to the limit $\sqrt{5}$.
- Hints:
 - Take the limit of both sides of the recurrence relation.
 - Use some reasonable assumptions about how limits should behave.
 - Deduce that our sequence converges to $\sqrt{5}$. You will need to argue carefully about the positive or negative square root.

- **Task 7.**
- Now use our recurrence relation to generate a sequence of better and better rational approximations to $\sqrt{5}$.
- Hint:
 - program the recurrence relation into a Python loop.